

ENGR 102 Summer Bridge Day 10 Instructor Key

Lectures 1-5 Recap, Direct Input, and Repetition Launch

20 points • approximately 20-25 minutes

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Boolean-operator convention. Python operators appear as [and](#), [or](#), and [not](#). Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Compute the stored values and exact output.

4 points

```
x = 17
y = 5
result = (x // y) ** 2 + x % y
ok = result >= 11 and not (x % 2 == 0)
print(result, ok)
```

Show the arithmetic state needed to support your final line.

Answer and explanation

17 // 5 is 3 and 17 % 5 is 2, so $\text{result} = 3 ** 2 + 2 = 11$. The [and](#) expression combines $\text{result} \geq 11$, which is True, with [not](#) False, which is True. Exact output: 11 True.

Question 2. Trace the ordered decision.

4 points

```
score = 88
missing = True

if score >= 85 and not missing:
    message = "distinction"
elif score >= 70 or not missing:
    message = "pass"
else:
    message = "incomplete"

print(message)
```

Name the first matching branch and explain why the higher label is not selected.

Answer and explanation

The first condition is False because missing is True, so [not](#) missing is False. The elif condition is True because $\text{score} \geq 70$ is True. Exact output: pass. Branch order does not make distinction run when its complete condition is False.

Question 3. Repair the Boolean rule.

4 points

```
# Intended: ready only when calibrated and 10 <= reading <= 40
ready = calibrated or reading >= 10 and reading <= 40
```

Write the corrected assignment and give one input that distinguishes the two versions.

Answer and explanation

Correct: `ready = calibrated and 10 <= reading <= 40`. Distinguishing test: `calibrated = False` and `reading = 25`. The buggy expression gives True from the range condition, while the corrected expression gives False.

Question 4. Program construction**8 points**

Write one complete Python program that satisfies every requirement.

- Read temperature as a float and calibrated as text, where exactly yes means calibrated.
- Print invalid when temperature is below -20 or above 60.
- Otherwise print ready only when calibrated is yes and temperature is from 15 through 35 inclusive; print hold for every other valid case.
- Use one if/elif/else chain and print exactly one message.

Answer and explanation

```
temperature = float(input())
calibrated = input()

if temperature < -20 or temperature > 60:
    print("invalid")
elif calibrated == "yes" and 15 <= temperature <= 35:
    print("ready")
else:
    print("hold")
```

The invalid range must be checked first. The ready branch then combines calibration with the inclusive operating range. The final else collects all valid but not-ready cases.

Protective tests include -20/yes -> hold, 15/yes -> ready, 35/yes -> ready, 36/yes -> hold, and 61/yes -> invalid.

Scoring guidance

2 input/conversion, 2 invalid boundary, 2 combined ready condition, 1 mutually exclusive branch structure, 1 exact single message.